Boris's Brexit Dilemma as a Game

Players, tree, subgame-perfect equilibrium and the value of commitment

Case write-up for Lecture 3 — Economics of Organizations and Incentives (MPE Insper)

November 2026

The assignment and the sources

Cases. "Boris's Brexit Dilemma (A)" and "Boris's Brexit Dilemma (B)" (Darden, UV9621 and UV9624). The case text is not reproduced here. The model uses only the cost figures summarized in §4; every other element (the political cost r, the deterrence premium k, the variants) is an addition of this write-up and is labelled as such.

Assignment (`Cases/Brexit/intrucoes.md`, quoted in the original Portuguese). In pairs, students negotiate a Brexit deal with their counterpart and write the negotiation as a game; in class each side decides whether to prepare for a no-deal Brexit (UK), which deal to offer (EU) and whether to take the EU's proposal to Parliament (UK). The task list:

> Para modelar a negociação como um jogo, seu grupo deve: (i) identificar os jogadores; (ii) indicar a ordem das jogadas e as ações disponíveis para cada jogador em cada ponto de decisão; (iii) indicar o que cada jogador sabe quando joga (por exemplo, a UE observa se o Reino Unido se preparou para um Brexit sem acordo?); (iv) atribuir payoffs a cada resultado final com base nas informações do caso (os números exatos importam menos do que uma ordenação que vocês consigam justificar); (v) desenhar a árvore do jogo; (vi) resolver o jogo por indução retroativa e encontrar o equilíbrio perfeito em subjogos; e (vii) comparar o equilíbrio com o resultado da negociação de vocês e, se forem diferentes, explicar por quê.

Method. Extensive form, strategies, backward induction and subgame perfection: Tadelis (2013) Ch. 7 (pp. 129–150) and Ch. 8 (pp. 151–174); credibility and commitment: Cabral (2017) §7.2, as in `Notas/L03/L03-06-credibility-entry-commitment.md`. Every number below is asserted with exact fractions by `Cases/Brexit/brexit-check.py`, which also runs backward induction programmatically on the tree.

Executive summary

Subgame-perfect equilibrium (baseline: the EU sees whether the UK prepared; $r = 0$):

- **UK:** *prepare* for no deal; if prepared, present only the favourable deal and reject the neutral and unfavourable ones; if unprepared, present every deal.
- **EU:** offer the *favourable* deal if the UK prepared and the *unfavourable* deal if it did not.

On the equilibrium path the UK prepares, the EU offers the favourable deal and the UK presents it. Expected payoffs are $(-24.7, -14)$ in £bn, against $(-28.6, -7.6)$ if the UK had not prepared.

Lesson. A sunk, visible investment makes the threat to reject credible and shifts the EU's offer. Preparation lowers the UK's ex-post cost of walking away from 35 to 22.5, below the neutral and unfavourable deals. The EU, seeing this, concedes the favourable deal. The UK gains 3.9; the EU loses 6.4; the 2.5 difference is a deadweight loss.

Robustness. The result needs the preparation to be *observed*: if it is hidden, there is no pure equilibrium, and the UK prepares with probability 8/13 while the EU offers the favourable deal with probability 11/36. Talks then break down with probability $50/117 \approx 0.43$. It also needs the political cost of walking away to be small ($r < 0.4$) and the EU's deterrence premium on a favourable deal to be moderate ($k \leq 5$).

Notation.

- $d \in \{F, Nt, U\}$: the soft-Brexit deal the EU offers (favourable, neutral, unfavourable to the UK); c_d and e_d are the UK's and the EU's costs of deal d.
- $H_{UK} = 35$, $H_{EU} = 18$: the costs of a hard (no-deal) Brexit.
- $\kappa = 5$: the UK's preparation cost; $m = 12.5$: the hard-Brexit cost it saves.
- $p = 0.8$: the probability that Parliament passes a presented deal.
- $r \geq 0$: the UK government's political cost of rejecting the EU's offer outright (not in the case's numbers). $k \geq 0$: the EU's deterrence premium on a favourable deal (not in the case's numbers).
- Payoff = −cost, in £bn. Players are risk neutral: they rank lotteries by expected cost.

1 (i) The players

- **The United Kingdom**, represented by the government of Boris Johnson. It moves twice: it chooses whether to prepare for no deal, and whether to take the EU's offer to Parliament.
- **The European Union**, represented by its chief negotiator, Michel Barnier. It chooses which deal to offer.
- **Parliament, modelled as Nature.** A presented deal passes with probability $p = 0.8$ and fails with probability 0.2; a failed vote means a hard Brexit.

Why Parliament is Nature. The case data give the vote as a probability, not as the choice of an agent with stated payoffs. The government cannot control it, and the EU cannot bargain with it directly. A chance move with known probabilities is the standard way to put such a step into an extensive form (Tadelis §7.1, Definition 7.1, the "Nature" player). The model also makes p independent of the deal type, which is what a Nature move with one probability implies.

Remark: Parliament as a strategic player. A strategic Parliament (say, a pivotal MP who compares each deal with no deal) would make three changes. (1) The game has three players, and the vote becomes a best response, so p depends on the deal: an unfavourable deal is more likely to fail. §7.4 shows that this *helps* the UK, because the EU then has a reason to offer

more even to an unprepared UK. (2) The government may want a *tougher* Parliament: a veto player that rejects bad deals is a commitment device. This is Schelling's (1960) point that the power to bind oneself is bargaining power, and Putnam's (1988) "two-level game". (3) With several parties in Parliament, the vote is itself a bargaining game; the legislative bargaining of `Notas/L04/L04-08-legislative-bargaining.md` would be the tool. We keep Parliament as Nature in the baseline because it keeps the tree solvable by hand and isolates the commitment effect.

2 (ii) Timing and actions

Stage 1. **UK** chooses *Prepare* (P) or *Not prepare* (N) for a no-deal Brexit. Preparation costs $\kappa = 5$, is all-or-nothing and is sunk: it is paid even if a deal is later implemented.

Stage 2. **EU** offers a soft-Brexit deal $d \in \{F, Nt, U\}$.

Stage 3. **UK** either *presents* the deal to Parliament or *rejects* it outright. Rejection means a hard Brexit, plus the political cost r.

Stage 4. **Nature (Parliament)** passes a presented deal with probability 0.8 (the deal is implemented) or rejects it with probability 0.2 (hard Brexit, no political cost r: the government tried).

Strategies. A pure strategy names an action at every decision node, including nodes that are never reached (Tadelis Definition 7.4, pp. 138–139). The UK has one root node and six present/reject nodes, one for each (preparation, offer) pair. The EU has two nodes, one for each preparation choice. Counting by Tadelis's rule $|S_i| = \prod_k m_k$:

$$|S_{UK}| = 2 \times 2^6 = 128, \qquad |S_{EU}| = 3^2 = 9.$$

A UK strategy therefore reads like "P; after P: present F, reject Nt, reject U; after N: present all three". The EU's reads like "F after P, U after N".

3 (iii) Information: who knows what

Baseline: the EU observes preparation. Stockpiling, contingency contracts and border infrastructure are visible, so the EU knows whether the UK prepared when it makes its offer. The UK sees the offer before deciding whether to present it. Every information set is a singleton. Because Nature moves, Tadelis's Definition 7.3 (p. 136) formally classifies the game as one of imperfect information. Still, no player is ever unsure about an earlier move: Nature moves last, and its probability is common knowledge. The UK presents without knowing how the vote will go, but that is risk, not private information. We also take r and k as common knowledge in the baseline; §11 discusses what changes when they are not.

Variant: the EU does not observe preparation. The EU's two nodes (after P and after N) lie in one information set, drawn as a dashed line in Figure 2. The EU must choose one offer for both. The UK still knows its own preparation and sees the offer, so its six present/reject nodes remain singletons (perfect recall, Tadelis Definition 7.7, p. 142). The EU's strategy set shrinks to $\{F, Nt, U\}$. Strategically, stage 1 and stage 2 become simultaneous: the same tree with a different information structure, as in the sequential and simultaneous Battle of the Sexes of `Notas/L03/L03-01-extensive-form.md` §4. §8 solves this variant.

4 (iv) Payoffs

4.1 Inputs

Outcome (costs in £bn)	UK	EU
Hard Brexit, UK unprepared	35	18
Soft Brexit, favourable deal F	19	13
Soft Brexit, neutral deal Nt	23	9
Soft Brexit, unfavourable deal U	27	5
Preparation: cost κ, hard-Brexit cost saved m	5; 12.5	—
Probability Parliament passes a presented deal	$p = 0.8$	

Preparation saves 2.5 per pound spent ($12.5 = 2.5 \times 5$), but only in a hard Brexit. In a soft Brexit the 5 is lost. The EU's costs do not depend on whether the UK prepared.

4.2 How each terminal payoff is computed

A prepared UK's hard-Brexit cost is the unprepared cost, minus the saving, plus the sunk outlay:

$$\begin{aligned} H_{UK} - m + \kappa &= 35 - 12.5 + 5 \\ &= 22.5 + 5 \\ &= 27.5. \end{aligned}$$

A prepared UK in a soft Brexit pays the deal plus the outlay, $c_d + 5$. A rejection adds r to the UK's hard-Brexit cost. The eighteen terminal nodes therefore carry only eight distinct payoff pairs (Table 1).

Table 1: Terminal payoffs (UK, EU) in £bn. "Fail" is the vote failing; "reject" is the UK rejecting outright.

		Offer d		
UK prepared?	Event	F	Nt	U
N	deal passes	$(-19, -13)$	$(-23, -9)$	$(-27, -5)$
N	vote fails	$(-35, -18)$	$(-35, -18)$	$(-35, -18)$
N	UK rejects	$(-35 - r, -18)$	$(-35 - r, -18)$	$(-35 - r, -18)$
P	deal passes	$(-24, -13)$	$(-28, -9)$	$(-32, -5)$
P	vote fails	$(-27.5, -18)$	$(-27.5, -18)$	$(-27.5, -18)$
P	UK rejects	$(-27.5 - r, -18)$	$(-27.5 - r, -18)$	$(-27.5 - r, -18)$

4.3 Why the ordering is defensible

The assignment stresses that a justified ordering matters more than exact numbers. Four ordinal facts drive the whole solution; any numbers that preserve them give the same equilibrium.

(O1) **An unprepared UK prefers every deal to no deal:** $27 < 35$. Even the unfavourable deal avoids the worst disruption.

(O2) **A prepared UK prefers no deal to the neutral and unfavourable deals, but not to the favourable one:** $19 < 22.5 < 23 < 27$. Here 22.5 is the prepared hard-Brexit cost net of the sunk 5, which is paid in either case.

(O3) **The EU prefers any implemented deal to no deal, and less generous deals to more generous ones:** $5 < 9 < 13 < 18$. A hard Brexit hurts the EU too, but less than the UK ($18 < 35$): the EU is the larger party and the one with the better outside option.

(O4) **Preparation plus the favourable deal beats no preparation plus the unfavourable deal for the UK**, in expectation: $24.7 < 28.6$ (computed in §6).

The complete ranking of the UK's sure outcomes is $(N, F) \succ (N, Nt) \succ (P, F) \succ (N, U) \succ (P, \text{hard}) \succ (P, Nt) \succ (P, U) \succ (N, \text{hard})$, that is, $-19 > -23 > -24 > -27 > -27.5 > -28 > -32 > -35$. The EU ranks $U \succ Nt \succ F \succ$ hard whatever the UK did.

Every deal costs the two sides 32 jointly ($19 + 13 = 23 + 9 = 27 + 5$). The deal type is therefore a pure transfer between the two sides; only the probability of a hard Brexit and the preparation outlay change total costs (§9).

4.4 What the payoffs leave out

- **The political cost** r of walking away from an offer. A government that refuses to put a deal to a vote may be blamed for the hard Brexit that follows; one that loses a vote is less exposed. The baseline sets $r = 0$, and §7.1 shows that it matters a lot.
- **The EU's deterrence motive.** The EU wants low costs, but it also wants to deter other members from leaving. A generous deal for a leaver sets a precedent. We represent this as a premium k added to the EU's cost of an implemented favourable deal (§7.3).
- **Non-monetary values**: sovereignty, the Irish border and the peace process, the integrity of the Union and of the single market, careers and party politics. They enter only through r and k, or through the ordering.
- **Risk attitudes, dynamics and uncertainty**: risk neutrality is assumed; the future relationship and later negotiations are absent; the costs, p and r are known.

5 (v) The game tree

Figure 1 is the full tree of the baseline game. Filled dots are decision nodes, open dots are Nature's (Parliament's) moves; the payoffs are (UK, EU). Thick green edges are the subgame-perfect choices at *every* node, on and off the equilibrium path (§6).

6 (vi) Backward induction and the subgame-perfect equilibrium

The game is finite and every information set is a singleton. Zermelo's argument (Tadelis Proposition 8.1) carries over once each chance node is replaced by its expected payoffs, so backward induction yields a subgame-perfect equilibrium. Every decision node starts a subgame, so the backward-induction profile is the SPE (Tadelis §8.2). With $r = 0$ there are no ties, so the SPE is unique. Throughout, the UK and the EU compare *expected costs*.

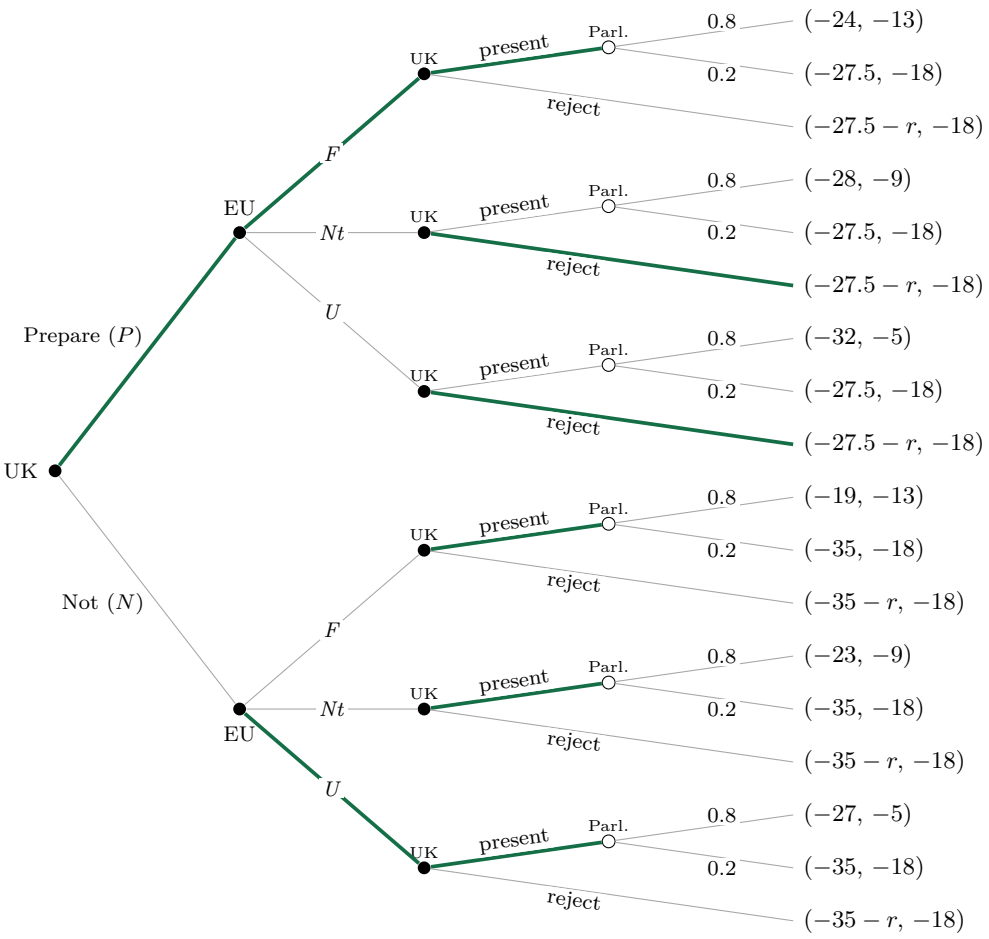


Figure 1: The baseline game (the EU observes preparation). At each Parliament node the upper edge is "pass" (0.8) and the lower is "fail" (0.2). Green: the SPE action at every node, for $r < 0.4$ and $k \leq 5$.

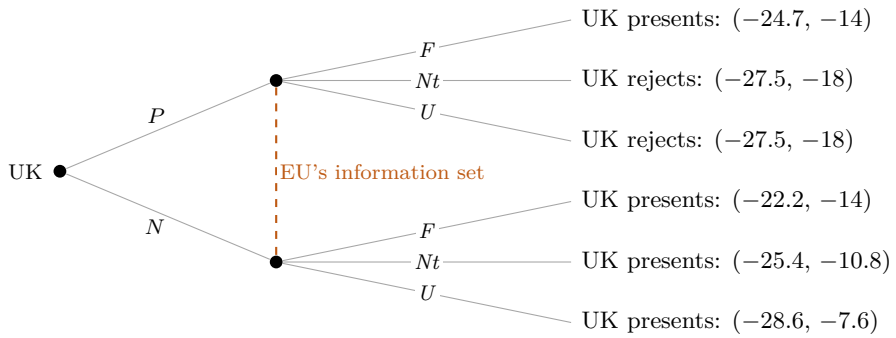


Figure 2: The unobservable variant, reduced. The EU's two nodes are one information set. After each offer the UK still moves at a singleton node, so each leaf shows the UK's optimal present/reject choice (from Figure 1, $r = 0$) and the expected payoffs it yields.

6.1 Step 1: the UK's present/reject decisions (six nodes)

After N. Presenting deal d costs $0.8\,c_d + 0.2 \times 35$; rejecting costs $35 + r$. For the favourable deal:

$$\begin{aligned} 0.8 \times 19 + 0.2 \times 35 &= 15.2 + 7 \\ &= 22.2. \end{aligned}$$

In the same way, Nt costs $18.4 + 7 = 25.4$ and U costs $21.6 + 7 = 28.6$. All three are below 35, and therefore below $35 + r$ for any $r \geq 0$. **An unprepared UK presents every deal.**

After P. Presenting deal d costs $0.8\,(c_d + 5) + 0.2 \times 27.5$; rejecting costs $27.5 + r$. For the favourable deal:

$$\begin{aligned} 0.8 \times 24 + 0.2 \times 27.5 &= 19.2 + 5.5 \\ &= 24.7. \end{aligned}$$

In the same way, Nt costs $22.4 + 5.5 = 27.9$ and U costs $25.6 + 5.5 = 31.1$. With $r = 0$ only $24.7 < 27.5$: **a prepared UK presents F and rejects Nt and U.**

The general present/reject rule. For any $p \in (0, 1]$ and $r \geq 0$, a prepared UK presents d when the expected cost of presenting does not exceed that of rejecting:

$$\begin{aligned} 5 + p\,c_d + (1 - p)\,22.5 &\leq 27.5 + r \\ p\,c_d + 22.5 - 22.5\,p &\leq 22.5 + r \\ p\,(c_d - 22.5) &\leq r \\ c_d &\leq 22.5 + \frac{r}{p}. \end{aligned} \tag{1}$$

The sunk 5 is paid in both cases, so it drops out of (1). What moves the threshold is the *saving* $m = 12.5$, which lowers the ex-post cost of no deal from 35 to 22.5. For an unprepared UK the same steps give $c_d \leq 35 + r/p$, which every deal satisfies.

Table 2: Step 1: the UK's expected cost of presenting each deal versus rejecting it ($p = 0.8$).

UK at node	present F	present Nt	present U	reject
after N	**22.2**	**25.4**	**28.6**	$35 + r$
after P	**24.7**	27.9	31.1	**27.5** + **r** (for Nt, U)

6.2 Step 2: the EU's offer (two nodes)

If a deal is presented, the EU's expected cost is $0.8\,e_d + 0.2 \times 18 = 0.8\,e_d + 3.6$:

$$F : 0.8 \times 13 + 3.6 = 14, \qquad Nt : 0.8 \times 9 + 3.6 = 10.8, \qquad U : 0.8 \times 5 + 3.6 = 7.6.$$

If the UK rejects, the EU's cost is 18.

- **After** N every deal is presented, so the EU picks the cheapest for itself: U (cost 7.6). The UK's cost is 28.6.
- **After** P only F is presented. The EU compares F (14) with Nt or U, which are rejected (18), and offers F. The UK's cost is 24.7.

6.3 Step 3: the UK's preparation decision

Preparing leads to $(-24.7,\ -14)$ and not preparing to $(-28.6,\ -7.6)$. Since $24.7 < 28.6$, the UK **prepares**. Its gain is $28.6 - 24.7 = 3.9$, and the EU's loss is $14 - 7.6 = 6.4$.

Answer. The subgame-perfect equilibrium (baseline, $r = 0$, $k = 0$) is the strategy profile

$$s^*_{UK} = (P;\ \text{after } P\text{: present } F,\ \text{reject } Nt,\ \text{reject } U;\ \text{after } N\text{: present } F, Nt, U),$$
$$s^*_{EU} = (F \text{ after } P;\ U \text{ after } N),$$

with Parliament passing a presented deal with probability 0.8. On the equilibrium path the UK prepares, the EU offers F, and the UK presents it. With probability 0.8 the vote passes, giving $(-24,\ -13)$. With probability 0.2 it fails, giving a prepared hard Brexit, $(-27.5,\ -18)$. Expected payoffs are $(-24.7,\ -14)$. The actions after N are off the path, yet they are part of the equilibrium: they tell the UK what not preparing would cost it.

6.4 A Nash equilibrium that is not subgame perfect

Consider the profile in which the UK does *not* prepare but says it would reject anything but the favourable deal, and the EU believes it:

$$s'_{UK} = (N;\ \text{after } N\text{: present } F,\ \text{reject } Nt, U;\ \text{after } P\text{: present } F,\ \text{reject } Nt, U),$$
$$s'_{EU} = (F \text{ after } P;\ F \text{ after } N).$$

The outcome is N, then F, presented, with expected payoffs $(-22.2,\ -14)$. This is better for the UK than the SPE.

It is a Nash equilibrium. Given s'_{EU}, preparing would cost the UK 24.7 instead of 22.2, and presenting F is optimal at the node that is reached. Given s'_{UK}, offering Nt or U after N would be rejected and cost the EU 18 instead of 14.

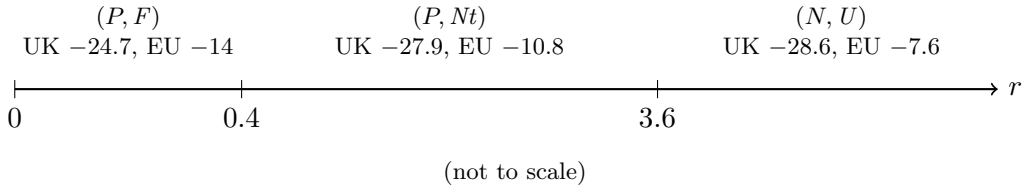
It is not subgame perfect. In the subgame after (N, U), rejecting costs the UK 35 while presenting costs 28.6. The threat to reject is never tested on the path, so Nash equilibrium does not discipline it; subgame perfection does (Tadelis §8.2, `Notas/L03/L03-05-subgame-perfection.md`). Without preparation the threat is cheap talk.

The UK's *price of credibility* is the gap between this non-credible ideal and the SPE: $24.7 - 22.2 = 2.5 = 5 - 0.2 \times 12.5$. That is the outlay minus its expected saving. The script enumerates all 128×9 pure-strategy profiles: there are 88 pure Nash equilibria, with outcomes $(-22.2, -14)$, $(-24.7, -14)$, $(-25.4, -10.8)$ and $(-27.5, -18)$, and exactly **one** of them is subgame perfect. As in the ultimatum game (Tadelis Proposition 11.1), Nash equilibrium "has no bite" here. Non-credible threats support almost anything.

7 Sensitivity

7.1 The political cost of walking away, r

By (1) with $p = 0.8$, a prepared UK presents Nt iff $23 \le 22.5 + r/0.8$, that is, $r \ge 0.8 \times 0.5 = 0.4$. It presents U iff $27 \le 22.5 + r/0.8$, that is, $r \ge 0.8 \times 4.5 = 3.6$. The after-$N$ rule never changes. Three regimes follow (ties at the thresholds are broken toward presenting):



(not to scale)

- $r < 0.4$: the baseline. A prepared UK presents only F; the EU offers F.
- $0.4 < r < 3.6$: a prepared UK presents F and Nt, so the EU offers Nt (10.8 beats 14). Preparing gives the UK $5 + 0.8 \times 23 + 0.2 \times 22.5 = 27.9$, which still beats 28.6, so the UK prepares. Preparation buys it only the neutral deal, a gain of 0.7.
- $r > 3.6$: even a prepared UK would present U, so the EU offers U in both subgames. Preparing would cost $5 + 0.8 \times 27 + 0.2 \times 22.5 = 31.1 > 28.6$, so the UK does not prepare. The outcome is (N, U).

The comparative static is the opposite of what one might expect. A higher political cost of walking away is a cost of *carrying out* the threat, so it weakens the threat. The UK wants a low r, for instance a public that would not blame it for no deal. A *negative* r, a political reward for walking away, would make the threat credible even without preparation: an unprepared UK rejects U iff $28.6 > 35 + r$, that is, iff $r < -6.4$.

7.2 A general pass probability p

With $r = 0$, rule (1) gives "present F only" after P for every $p > 0$. After N the UK presents every deal and the EU offers U. The UK's gain from preparing is

$$\begin{aligned}[27p + 35(1-p)] - [5 + 19p + 22.5(1-p)] &= (27 - 19 - 5)\,p + (35 - 22.5 - 5)(1-p) \\ &= 3p + 7.5(1-p) \\ &= 7.5 - 4.5p,\end{aligned}$$

which is positive for every $p \in [0, 1]$ and equals 3.9 at $p = 0.8$. The two terms read naturally. If the deal passes, preparation buys a better deal ($27 - 19 = 8$) for 5. If it fails, preparation is insurance ($35 - 22.5 = 12.5$) for the same 5.

For general p the thresholds in r become $r \geq 0.5\,p$ (present Nt) and $r \geq 4.5\,p$ (present U). In the middle regime the UK prepares iff $27.5 + 0.5p < 35 - 8p$, that is, $p < 15/17 \approx 0.88$. In the high regime preparation is pure insurance, and the UK prepares iff $27.5 + 4.5p < 35 - 8p$, that is, $p < 0.6$. A Parliament likely to pass deals (p near 1) makes insurance worthless and leaves only the strategic motive.

7.3 The EU's deterrence premium k

Suppose an implemented favourable deal costs the EU $13 + k$, where k is the reputational cost of rewarding a leaver in the eyes of other members. After P, the EU offers F iff

$$\begin{aligned}0.8\,(13 + k) + 0.2 \times 18 &\leq 18 \\ 10.4 + 0.8k + 3.6 &\leq 18 \\ 0.8k &\leq 4 \\ k &\leq 5.\end{aligned}$$

For $k > 5$ the EU offers Nt or U to a prepared UK, both are rejected, and the game ends in a hard Brexit: $(-27.5, -18)$. The UK still prepares ($27.5 < 28.6$), so the outcome is no deal by

design, with joint cost 45.5. (If the premium were paid on *offering* F rather than on implementing it, the condition would be $14 + k \leq 18$, that is, $k \leq 4$.) After N the EU offers U whatever k is. Deterrence concerns can thus turn a commitment that works into a breakdown: a tough EU and a credible UK are a bad combination.

7.4 Pass probabilities that depend on the deal

Let p_d be the probability that Parliament passes deal d. Relative to a hard Brexit, a presented deal saves the EU $p_d\,(18 - e_d)$: $5p_F$, $9p_{Nt}$ and $13p_U$. An unprepared UK still presents everything, so the EU offers U to it only if $13p_U \geq \max\{9p_{Nt},\, 5p_F\}$. Take $(p_F, p_{Nt}, p_U) = (0.9,\, 0.8,\, 0.5)$. The savings are 4.5, 7.2 and 6.5, so the EU offers Nt after N and the UK's cost is 25.4. After P, with $r = 0$, rule (1) (with p_d) still presents only F, so the EU offers F: the EU's cost is $0.9 \times 13 + 0.1 \times 18 = 13.5$ and the UK's is $5 + 0.9 \times 19 + 0.1 \times 22.5 = 24.35$. The UK still prepares, but its gain falls to 1.05. A Parliament that is hard on bad deals does part of the work of preparation.

8 The unobservable variant: commitment must be seen

If the EU cannot see preparation, the only proper subgames start at the UK's present/reject nodes. Those decisions are as in Step 1. Substituting them gives the reduced normal form (rows: UK; columns: EU; payoffs (UK, EU)):

	F	Nt	U
P	$(-24.7,\ \mathbf{-14})$	$(\mathbf{-27.5},\ -18)$	$(\mathbf{-27.5},\ -18)$
N	$(\mathbf{-22.2},\ -14)$	$(\mathbf{-25.4},\ -10.8)$	$(-28.6,\ \mathbf{-7.6})$

Bold marks best responses: the EU's within a row, the UK's within a column (for Nt, the UK's best response is N; after P, Nt and U tie for the EU at -18, but F beats both).

No pure Nash equilibrium. The best responses cycle. Against P the EU offers F; against F the UK does not prepare; against N the EU offers U; against U the UK prepares. This is the structure of Matching Pennies (`Notas/L02/L02-07-mixed-strategies.md`) and of the Lecture 3 inspection mini case "Work or shirk?". Preparation only pays if the EU expects it and concedes; but if the EU expects it, the UK would rather save the 5.

The mixed equilibrium. Let $\sigma = \Pr(P)$ and $\tau = \Pr(F)$. Nt is never used: it costs the EU at least as much as U in each row, and strictly more against N. The EU is indifferent between F and U when

$$\begin{aligned} -14 &= \sigma\,(-18) + (1-\sigma)(-7.6) \\ 14 - 7.6 &= (18 - 7.6)\,\sigma \\ \sigma &= \frac{6.4}{10.4} = \frac{8}{13}. \end{aligned}$$

The UK is indifferent between P and N when

$$\begin{aligned} -24.7\,\tau - 27.5\,(1-\tau) &= -22.2\,\tau - 28.6\,(1-\tau) \\ 28.6 - 27.5 &= (28.6 - 27.5 + 24.7 - 22.2)\,\tau \\ \tau &= \frac{1.1}{3.6} = \frac{11}{36}. \end{aligned}$$

At $\sigma = 8/13$, Nt would cost the EU $\frac{8}{13}\,18 + \frac{5}{13}\,10.8 = 198/13 \approx 15.2 > 14$, so it stays unused. Equilibrium payoffs:

$$u_{UK} = -27.5 + 2.8\,\tau = -27.5 + \tfrac{2.8\times 11}{36} = -\tfrac{1199}{45} \approx -26.64, \qquad u_{EU} = -14.$$

Lessons. (1) Hidden commitment is no commitment. With observability the UK's cost is 24.7; without it, 26.64. Being seen to prepare is worth $35/18 \approx 1.94$ to the UK. The EU's payoff, -14, is the same in both games. (2) Breakdown becomes likely. A prepared UK meets the unfavourable offer and rejects it with probability $\sigma(1-\tau) = 50/117 \approx 0.43$. The overall probability of a hard Brexit rises from 0.2 to $317/585 \approx 0.54$. (3) The UK therefore wants its preparation to be verifiable, and the EU, which gets -14 either way, has no reason to block that. Lecture 3's point that a commitment must be observed and irreversible (`Notas/L03/L03-06-credibility-entry-commitment.md` §3) is exactly this.

9 Efficiency: preparation as rent-shifting

Outcome	UK cost	EU cost	Joint cost
(N, U): no preparation ($r > 3.6$, or the benchmark)	28.6	7.6	36.2
(P, F): SPE, $r < 0.4$	24.7	14	38.7
(P, Nt): SPE, $0.4 < r < 3.6$	27.9	10.8	38.7
Unobservable preparation, mixed equilibrium	26.64	14	40.64
(P, hard): $k > 5$	27.5	18	45.5

Every deal costs 32 jointly, and a hard Brexit costs 53 jointly (40.5 if the UK prepared, before the outlay). So

$$(N, U) : 0.8 \times 32 + 0.2 \times 53 = 25.6 + 10.6 = 36.2,$$
$$(P, F) : 5 + 0.8 \times 32 + 0.2 \times 40.5 = 5 + 25.6 + 8.1 = 38.7.$$

The difference, $2.5 = 5 - 0.2 \times 12.5$, is the outlay minus its expected saving. Preparation is socially worthwhile only if $(1 - p)\,12.5 \geq 5$, that is, $p \leq 0.6$. At $p = 0.8$ it is a **rent-shifting investment with a deadweight loss**: it moves 6.4 from the EU to the UK and destroys 2.5 along the way $(6.4 - 3.9 = 2.5)$.

This is the logic of strategic investment in industrial organization. An incumbent sinks capacity, or posts a bond, not because the investment is productive but because it changes what the rival does next (Cabral §7.2, `Notas/L03/L03-06-credibility-entry-commitment.md`; the bond there is worth 30 to the incumbent). There is one difference. Cabral's bond costs nothing on the equilibrium path, whereas here the 5 is spent even when the deal passes, which is where the deadweight loss comes from.

A bargain the parties cannot strike. Suppose a deal on the same frontier ($c + e = 32$) gave the UK $c^* = (24.7 - 0.2 \times 35)/0.8 = 22.125$. An unprepared UK would then be exactly as well off as in the SPE, and the EU would pay $0.8 \times 9.875 + 3.6 = 11.5 < 14$. Both would gain from the EU promising "a generous deal if you do not prepare". But the promise is not credible: once the UK is unprepared, the EU's best reply is U. The inefficiency comes from a commitment problem on the EU's side. Moreover, no deal in the case's menu achieves it: Nt leaves an unprepared UK at 25.4.

10 In-class decision guide

Playing the UK

1. **Prepare, and make sure the EU sees it.** Visible preparation is worth 3.9 at $p = 0.8$ (and $3p + 7.5(1 - p) > 0$ for any p). Hidden preparation is worth far less (§8).
2. **If prepared: present only the favourable deal**, and reject the neutral and unfavourable ones, as long as your political cost of walking away is small ($r < 0.4$). If $0.4 < r < 3.6$, present F or Nt and reject U. If $r > 3.6$, do not prepare at all.
3. **If unprepared: present whatever is offered.** Every deal beats no deal. Saying otherwise is cheap talk, and a well-briefed EU will ignore it.
4. **Expect a breakdown if the EU's deterrence concern is large** ($k > 5$): you still prefer to have prepared ($27.5 < 28.6$).

Playing the EU

1. **If you see preparation, offer the favourable deal**, unless you believe the UK's political cost of walking away is high. If you are unsure, offer Nt instead of F only if $\Pr(r \geq 0.4) > 5/9$, and U instead of F only if $\Pr(r \geq 3.6) > 5/13$ (from $q \times 10.8 + (1 - q) \times 18 = 14$ and $q \times 7.6 + (1 - q) \times 18 = 14$).
2. **If you see no preparation, offer the unfavourable deal.** It will be presented.
3. **If you cannot tell**, the equilibrium has you offering F with probability $11/36$ and U otherwise. In practice, invest in finding out.
4. **Weigh precedent.** If rewarding a leaver costs you more than 5 in deterrence, refuse the favourable deal, and accept that a hard Brexit follows.

What a counterpart may do differently

A UK negotiator may skip preparation because it looks like bad faith, or threaten rejection without preparing. An EU negotiator may offer the neutral deal as a "fair" compromise, or call a bluff because it believes r is high. Each departure maps to a parameter above (r, k, p_d, beliefs) or to the behavioural reasons in §11. Record which one it was.

11 (vii) Comparing the equilibrium with your negotiation

The SPE is a benchmark for perfectly rational, money-maximizing players with common knowledge of the tree. Real pairs often end elsewhere. The usual reasons:

- **Fairness.** In ultimatum-game experiments, proposers typically offer 25–50% of the pie and responders usually reject offers below about 20–25% (Tadelis, Remark pp. 223–224, citing Camerer 2003; `Notas/L04/L04-05-ultimatum-game.md` §4). Here the "pie" is the surplus of a deal over an unprepared hard Brexit, $53 - 32 = 21$. The deals give the UK $16/21 \approx 76\%$ (F), $12/21 \approx 57\%$ (Nt) and $8/21 \approx 38\%$ (U). By lab standards even U is not an insulting offer, so a fairness-motivated rejection of U by an unprepared UK would have to come from political humiliation (a negative component of r), not from the split itself. Conversely, an EU player may refuse F as "unfair" to the EU.

- **Reputation and repeated interaction.** The EU deals with 27 members and future applicants, so a concession to a leaver is read by others. This is the k of §7.3, and the repeated-game logic of `Notas/L04/L04-02-carrot-and-stick.md`. The UK and the EU also negotiate again (trade, security, fisheries), which may soften both sides.
- **Uncertainty about p and r.** If the EU does not know r, its offer is a screen (thresholds 5/9 and 5/13 in §10); if the UK does not know how Parliament will vote, it may misjudge (1). With private information, rational bargaining can end in breakdown or delay (`Notas/L04/L04-09-bargaining-private-information.md`).
- **Cheap talk versus commitment.** Statements like "no deal is better than a bad deal" change nothing in the model unless backed by preparation or by a change in r. Pairs in which the EU believed an unprepared UK's threat played the non-credible Nash equilibrium of §6.4.
- **Deadline extensions.** The model assumes a rejection means a hard Brexit now. If a rejection can instead lead to an extension and a new round, the game becomes one of alternating offers (`Notas/L04/L04-07-rubinstein-infinite-horizon.md`). The split then depends on patience, not only on outside options, and preparation matters less.
- **Domestic politics.** Votes, party leadership and elections make p, r and even the UK's objective endogenous (§1 remark: two-level games).

Template: record your negotiation against the SPE.

Decision	SPE prediction (baseline)	What we did, and why (if different)
UK: prepare?	Prepare, visibly	
EU: what did it observe?	Preparation	
EU: offer	F (after P); U would follow N	
UK: present or reject	Present F	
Parliament (dice / draw)	Pass with prob. 0.8	
Final outcome, payoffs	$(-24, -13)$ or $(-27.5, -18)$; exp. $(-24.7, -14)$	
Beliefs about r, k, p	$r = 0$, $k = 0$, $p = 0.8$	
Main reason for any gap	—	

12 Historical note (general knowledge, not from the case)

The facts in this section are general knowledge and are not taken from the case. Boris Johnson became Prime Minister in July 2019. In October 2019 his government and the EU agreed a revised Withdrawal Agreement, which replaced the Irish "backstop" with the Protocol on Ireland/Northern Ireland. The Article 50 deadline had been extended three times during 2019. After the December 2019 general election, Parliament approved the Withdrawal Agreement, and the UK left the EU on 31 January 2020, entering a transition period that ended on 31 December 2020. Michel Barnier led the EU's side in both negotiations. The **EU–UK Trade and Cooperation Agreement**

was agreed on 24 December 2020, applied provisionally from 1 January 2021, and entered into force on 1 May 2021. It provides for zero tariffs and zero quotas on goods that meet its rules of origin, and covers much less of trade in services. Which of the model's deal types, if any, these agreements correspond to is a question for class discussion, not something the model settles.

References

- Tadelis, S. (2013). *Game Theory: An Introduction*. Princeton University Press. Ch. 7, "Preliminaries" of dynamic games (pp. 129–150): extensive form, Nature, information sets, strategies. Ch. 8, "Credibility and Sequential Rationality" (pp. 151–174): backward induction, Proposition 8.1, subgame perfection. Ch. 6 (pp. 101–114): mixed strategies. §11.1 (pp. 220–224): the ultimatum game and its experimental Remark.
- Cabral, L. (2017). *Introduction to Industrial Organization*, 2nd ed. MIT Press. §7.2 (pp. 167–171): credible commitment and entry deterrence.
- Schelling, T. C. (1960). *The Strategy of Conflict*. Harvard University Press.
- Putnam, R. D. (1988). Diplomacy and domestic politics: the logic of two-level games. *International Organization* 42(3), 427–460.
- Course notes (Lectures 2–4):
  - `Notas/L03/`: `L03-01-extensive-form.md`, `L03-02-strategies.md`, `L03-04-sequential-rationality-backward-induction.md`, `L03-05-subgame-perfection.md`, `L03-06-credibility-entry-commitment.md`;
  - `Notas/L02/`: `L02-07-mixed-strategies.md`;
  - `Notas/L04/`: `L04-02-carrot-and-stick.md`, `L04-05-ultimatum-game.md`, `L04-07-rubinstein-infinite-horizon.md`, `L04-08-legislative-bargaining.md`, `L04-09-bargaining-private-information.md`;
  - Lecture 3 mini case "Work or shirk?": `Aula/Lecture03.pdf`, slides 41–42.
- Cases: "Boris's Brexit Dilemma (A)" and "(B)", Darden UV9621 and UV9624 (figures only, not reproduced).
- Check script: `Cases/Brexit/brexit-check.py` (exact fractions; backward induction and normal-form enumeration on the tree).